Francesco Ferri

ASSOCIATE PROFESSOR · OFFSHORE RENEWABLE ENERGY

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AAU research profile · ORCID 0000-0003-4325-7482

PROFILE

Associate Professor working on offshore renewable energy, with a focus on wave energy converters, hydrodynamic modelling, experimental validation and control. My work connects numerical methods with laboratory measurements to improve device performance and reliability.

ACADEMIC AND PROFESSIONAL EXPERIENCE

Associate Professor | 2022 - present

Aalborg University · Department of the Built Environment

Research on wave, tidal and floating wind energy systems: hydrodynamic interactions, array layout optimisation, mooring design and experimental validation. Project management, funding proposals, software development and PhD teaching.

Assistant Professor | 2014 - 2022

Aalborg University

Development and optimisation of wave, tidal and floating wind energy devices.

Senior Researcher | 2018 - 2020

University of Campania (UNICAMPANIA)

Research on the reliability and reuse of end-of-life offshore platforms as artificial reefs, including active calcium carbonate deposition.

Process Engineer | 2010 - 2011

Bonatti Engineering

Process design for sour-crude handling, phase separation and storage tanks; P&IDs; for procurement and pre-commissioning.

Research Assistant | 2008 - 2009

Biochemistry and microbiology

Recovery of high-value antioxidants from olive mill wastewater and assessment of environmental impacts.

EDUCATION

PhD · Aalborg University | 2011 - 2014

Wave energy converters: hydrodynamics, model validation, hardware-in-the-loop testing, power take-off systems and control.

MSc, Chemical and Process Engineering · University of Bologna | 2005 - 2008

Thesis on wastewater valorisation for biofuels and compost.

BSc, Chemical Engineering · University of Bologna | 2002 - 2005

Thesis on CFD analysis of cavitation on Kaplan turbine blades.

Research & teaching

SELECTED PUBLICATIONS

2026 · The Use of Hydroelastic Structures in Wave Energy Converters

Fitzgerald, C. J., Marhadour, G., Ferri, F., Andersen, J., Thomsen, S. G., Folley, M. & Ringwood, J. V.

Journal of Marine Science and Engineering, 14, 1681. Publication ↗

2025 · Experimental investigation of wave-to-force modelling uncertainty for wave energy converters

Celesti, M. L., Papini, G., Pasta, E., Peña-Sanchez, Y., Mosquera, F. D., Ferri, F. & Faedo, N.

Mechanical Systems and Signal Processing, 226, 112323. Publication ↗

2025 · Extending Wave in situ Measurements Through Gaussian Process Regression: An Experimental Campaign

Gambarelli, L., Pasta, E., Ferri, F. & Giorgi, G.

International Journal of Offshore and Polar Engineering, 35(3), 244-251. Publication ↗

2025 · Wave propagation over a submerged bar: benchmarking of VoF, sigma transformation, and SPH numerical models against physical wave flume tests

Andersen, J., Eldrup, M. R., Ferri, F. & Verao Fernandez, G.

Discover Applied Sciences, 7, 448. Publication ↗

2021 · Experimental Implementation and Validation of a Broadband LTI Energy-Maximizing Control Strategy for the Wavestar Device

Garcia-Violini, D., Pena-Sanchez, Y., Faedo, N., Windt, C., Ferri, F. & Ringwood, J. V.

IEEE Transactions on Control Systems Technology, 29(6), 2609-2621. Publication ↗

2018 · Cost Optimization of Mooring Solutions for Large Floating Wave Energy Converters

Thomsen, J. B., Ferri, F., Kofoed, J. P. & Black, K.

Energies, 11, 159. Publication ↗

2014 · Balancing power output and structural fatigue of wave energy converters by means of control strategies

Ferri, F., Ambühl, S., Fischer, B. & Kofoed, J. P.

Energies, 7(4), 2246-2273. Publication ↗

Full publication record: vbn.aau.dk/en/persons/ffer/publications/

TEACHING AND INTERNATIONAL ACTIVITIES

MSc teaching: Fundamentals of Fluid Mechanics; Oceanography and Time Series Analysis; analogue signals, data acquisition and introductory control for Water and Environmental Engineering.

PhD teaching: Numerical and Experimental Modelling and Control of Wave Energy Converters.

WECCCOMP | 2016 - 2019. Collaboration with Maynooth University, NREL and Sandia National Laboratories on numerical benchmarking and experimental assessment of wave energy control algorithms.

TECHNICAL SKILLS AND LANGUAGES

Programming: Python, MATLAB/Simulink, C++, JavaScript, Vue.js, HTML/CSS.
Methods: hydrodynamic modelling, experimental validation, control, mooring and array optimisation.
Languages: Italian, English, Danish, Spanish and Portuguese.